Jay Desai

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Inspire profile

Professional Summary

PhD candidate in theoretical physics with strong expertise in quantitative modeling, statistical inference; and extensive experience developing analytical frameworks and high-performance numerical workflows in Python and C++. Thesis research explores how model-independent Standard Model Effective Field Theory (SMEFT) framework helps to understand physics beyond the standard model. Multiple papers published in highly reputed peer-reviewed journals and presented research at international conferences. Looking for roles in quantum computing, with an interest in improving quantum error correction, hybrid quantum-classical algorithms, and algorithmic innovation to tackle classically hard problems.

Education

- **Stony Brook University** 2020 – Present
  Ph.D., Physics (High Energy Theory and Phenomenology)
- **Stony Brook University** 2018 – 2020
  M.A., Physics
- **Indian Institute of Technology Roorkee** 2014 – 2018
  B.Tech., Electronics and Communication Engineering, Minor in Physics

Research Experience

- **Graduate Research Assistant, C.N. Yang Institute for Theoretical Physics** 2020 – Present
  Stony Brook University
  - Constructed dimension-eight operator basis for universal effective field theories, cross-checked using representation theory techniques to compute group invariants.
  - Built analytical and computational tools to extract constraints from experimental datasets.
  - Designed and implemented Monte Carlo simulation workflows for collider phenomenology.
  - Performed large-scale numerical computations and symbolic manipulations using Python, Mathematica, and C++.
  - Collaborated in multi-institutional research teams; co-authored multiple peer-reviewed publications in Physical Review D journal.
- **Research Assistant, Center for Nuclear Science** 2018-2020
  Stony Brook University
  - Experience with large-scale simulation packages and C++ based data analysis framework ROOT.
  - Applied statistical reweighting techniques to estimate uncertainty reduction and generate actionable insights for the proposed electron-ion collider.

Publications

All papers can be found on my INSPIRE and Google Scholar profile.

- *"Drell-Yan production in universal theories beyond dimension-six SMEFT"*
  arXiv:2503.19962 [hep-ph], Phys. Rev. D 112, 013009 (2025)
  Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz
- *"Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"*
  arXiv:2404.03720 [hep-ph], Phys. Rev. D 110 (2024) 033003
  Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia
- *"Impact of dimension-eight SMEFT operators in the EWPO and Triple Gauge Couplings analysis in Universal SMEFT"*
  arXiv:2304.03305 [hep-ph], Phys. Rev. D 107, 115013 (2023)
  Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz

Technical Skills

- **Programming**: Python, C, C++, Java, IBM Qiskit
- **Scientific Computing**: NumPy, Mathematica, MATLAB, FeynCalc
- **Simulation & Modeling**: MadGraph, Pythia, ROOT, DJANGOH
- **Quantitative Methods**: Statistical inference, multi-parameter optimization, high-dimensional modeling
- **Tools**: Linux, Version Control, LaTeX

Selected Research Highlights

- Developed model-independent quantitative frameworks for analyzing high-dimensional parameter spaces in effective field theories.
- Quantified parameter constraints using precision observable datasets and global fitting approaches.
- Presented research at international conferences including CERN workshops and national physics meetings.

Seminars/Conference Talks

- **Brookhaven National Lab (BNL) Seminar (invited), Brookhaven, December 11th 2025**
  "Universal Standard Model Effective Field Theory beyond dimension-six"
- **Higgs and Effective Field Theory (HEFT), CERN, June 3rd 2025**
  "Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"
- **LHC EFT WG meeting, (online), April 28th 2025**
  "Drell-Yan production in universal theories beyond dimension-six SMEFT"
- **DPF-Pheno 2024, University of Pittsburgh, May 13th 2024**
  "Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"
- **Pheno 2023, University of Pittsburgh, May 8th 2023**
  "A Phenomenological study of Higgs jets a muon collider"
- **Higgs 2022, Pisa, Italy, Nov 9th 2022(hybrid)**
  "A Phenomenological study of Higgs jets a muon collider"
- **EIC Yellow Report: semi-inclusive working group meeting**
  "Impact study of Kaon SIDIS at EIC on polarized strange quark distribution function"
- **APS April Meeting, 2020**
  "Strange quark polarized distribution function using SIDIS at EIC"

Awards

- Gerald Brown Prize (Research Travel Award), Stony Brook University
- Nationwide Education and Scholarship Test (Ranked 13th nationally)

Leadership & Teaching

- Instructor and Teaching Assistant for undergraduate physics courses.
- Journal Referee for JHEP.

Courses

- Quantum Programming, Quantum Mechanics I, Quantum Mechanics II, Quantum Field Theory I, Quantum Field Theory II, Advanced Quantum Field Theory, Group Theory, Statistical Mechanics.